

The `while` Loop: Continuing Until a Condition Changes

Sometimes we need to repeat a process until a specific condition is met, rather than for a predetermined number of iterations. The `while` loop is perfect for this scenario.

```
1. capital = 10000
2. target = 15000
3. annual_return = 0.12
4. years = 0
5.
6. while capital < target:
7.     capital = capital * (1 + annual_return)
8.     years += 1
9.
10. print(f"It takes {years} years to reach ${target:,.2f} with 12% annual returns")
11.
12. This example calculates how many years it would take to grow a trading account to a target
    value with consistent returns.
13. Here's a trading-specific example:
14. price = 100.0
15. days = 0
16. volatility = 0.015 # 1.5% daily price change
17. import random # For simulating random price movements
18.
19. while 90 < price < 110: # Continue until price breaks out of range
20.     # Simulate random price movement
21.     change = price * volatility * (random.random() * 2 - 1)
22.     price += change
23.     days += 1
24.
25. print(f"Price broke out of range after {days} days")
26.
```

This simulates how long a stock might trade within a certain price range before breaking out—useful for options traders who profit from predicting volatility patterns.